

FOGNET SENTINEL

Verified Safe-Corridor Cooperative Safety Network

SIH26007 • Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions

Core idea: do not only detect fog. Continuously check whether the vehicle has enough verified road ahead to stop safely.

SENSE VERICALCULATE SET SAFEWARN +
SAFE CORRIDOR STOPPING SPEED COORDINATE

This document is a synthesis of the six uploaded solution documents. Each major feature is tagged to its source document.

1. The solution in simple words

A mine truck cannot always see far enough through fog. Our system therefore uses radar as the main obstacle sensor, camera as supporting perception, roadside fog sensing, vehicle-to-vehicle hazard sharing, and a physics-based safety engine.

The system continuously creates a **Verified Safe Corridor**. It compares this distance with the truck's conservative stopping requirement. If the verified corridor becomes too short, the recommended safe speed is reduced and the driver is warned.

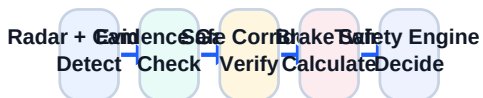
For blind sections, a **Virtual Block** and **Movement Token** prevent conflicting vehicles from entering the same dangerous section. Network information improves awareness, but local radar safety continues if communication fails.

Source: Snehal_NEXORA_FogNe.pdf • AboliFogNet_4.1.pdf • Chatgpt_FogNet_Nexus.pdf

One-line explanation for the team

“We verify how much road ahead is safe to trust, compare it with stopping distance, reduce safe speed when needed, and share hazards between vehicles.”

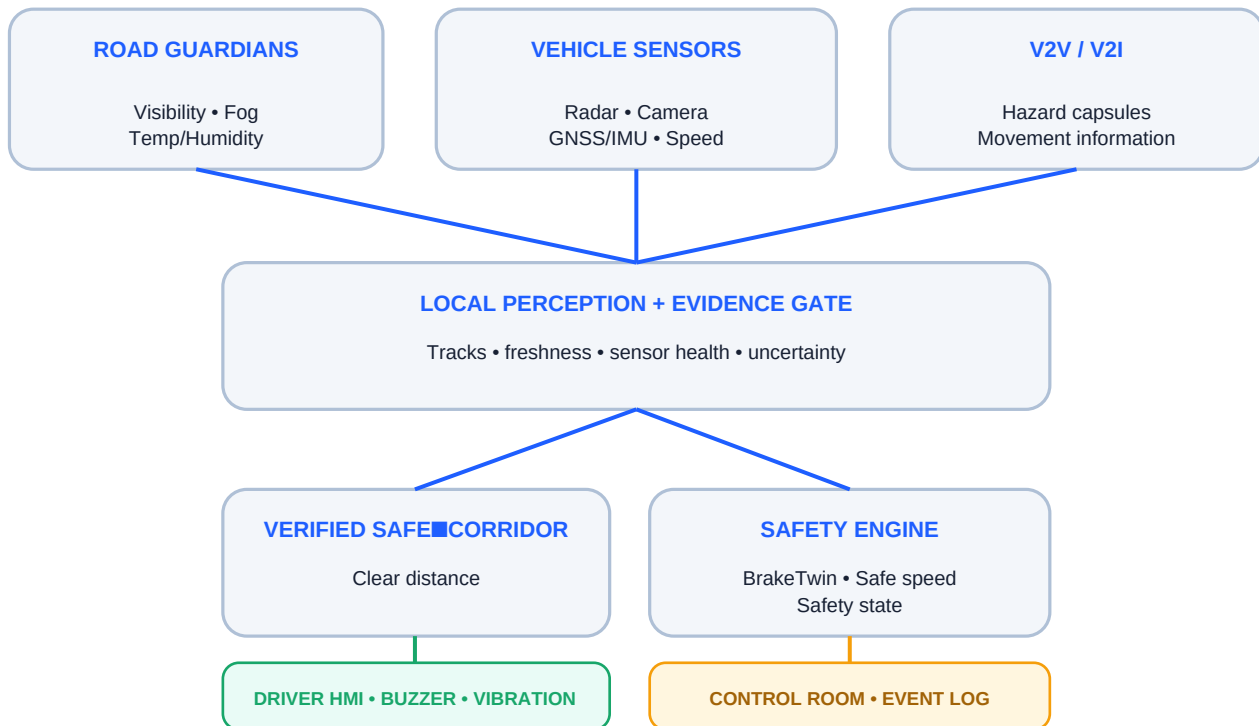
What each layer does



AI assists perception and analysis. The final safety boundary comes from deterministic, auditable safety logic.

Source: Snehal_NEXORA_FogNe.pdf • GLM_Fogswam.docx • PrathameshFOG_SHIELD_X.pdf

2. Complete system architecture

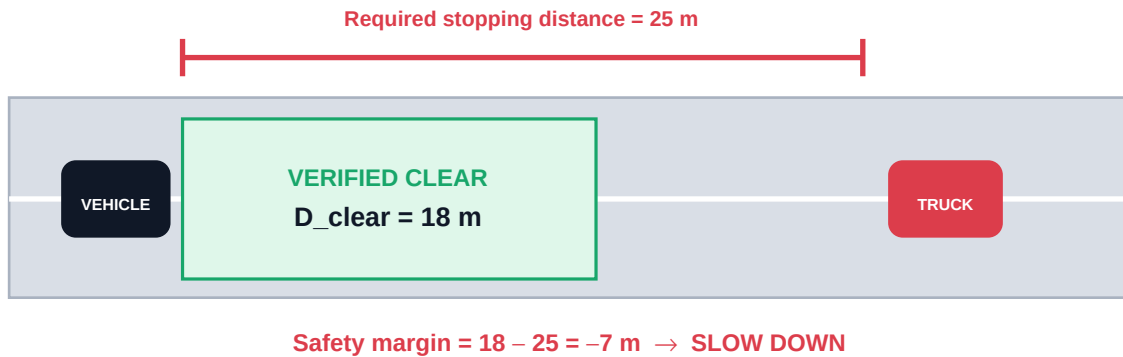


Safety hierarchy: local radar and local safety logic come first. Cooperative information, roadside information, and the control room add awareness but do not become the sole safety path.

Source: AboliFogNet_4.1.pdf • Snehal_NEXORA_FogNe.pdf • Chatgpt_FogNet_Nexus.pdf

3. Core innovation: Verified Safe Corridor

The system does not ask only “Is fog present?” It asks “How much of the road ahead can we currently verify as safe, and is that enough for this vehicle to stop?”



Example: if 18 m is verified clear but the vehicle needs 25 m to stop safely, the margin is –7 m. The system recommends slowing down before entering the unverified area.

Source: Snehal_NEXORA_FogNe.pdf • AboliFogNet_4.1.pdf

Required stopping distance

$$D_{\text{required}} = v \times t_{\text{reaction}} + \frac{v^2}{2 \times a_{\text{safe}}} + D_{\text{buffer}}$$

The actual implementation will use conservative, validated braking values and vehicle/road conditions.

Source: AboliFogNet_4.1.pdf • Snehal_NEXORA_FogNe.pdf

4. Vehicle sensing and cooperative awareness

Vehicle sensing



Source: Snehal_NEXORA_FogNe.pdf • PrathameshFOG_SHIELD_X.pdf • GLM_Fogswam.docx

Hazard Capsule

A detected hazard is shared as a small data message instead of streaming video. It can contain hazard type, relative distance, relative speed, route/segment, uncertainty, timestamp, evidence and expiry.

Important: peer information can add a hazard, but cannot silently clear a valid local radar hazard.

Source: GLM_Fogswam.docx • Snehal_NEXORA_FogNe.pdf

Example



Source: GLM_Fogswam.docx • PrathameshFOG_SHIELD_X.pdf

5. Blind curves and roadside fog intelligence

Virtual Block

A dangerous blind segment behaves like a controlled block. Vehicles reserve or occupy it so conflicting vehicles do not enter blindly from opposite directions.



Source: Snehal_NEXORA_FogNe.pdf

Movement Token

The traffic coordinator can grant a temporary movement token using vehicle position, speed, load, gradient, road width, stopping margin and section state. We do not assume that a loaded truck always gets priority.

Source: Chatgpt_FogNet_Nexus.pdf

Road Guardian

A roadside unit measures visibility/fog conditions and environmental variables. The system uses current visibility and a calibrated trend when site data supports it. Adaptive sampling can increase when conditions deteriorate.

Source: AboliFogNet_4.1.pdf • Chatgpt_FogNet_Nexus.pdf • Snehal_NEXORA_FogNe.pdf

6. AI, BrakeTwin and driver warning

AI is an assistant, not the final safety authority

AI can help with vehicle/object classification, tracking assistance, fog analysis, sensor anomaly detection and near-miss analysis. The deterministic safety engine decides the safe operating envelope.

Source: GLM_Fogswam.docx • PrathameshFOG_SHIELD_X.pdf • Snehal_NEXORA_FogNe.pdf

BrakeTwin

Stopping capability is treated conservatively using vehicle class, load, grade, surface condition and validated deceleration evidence. If evidence becomes worse, speed can be reduced immediately; it should not silently justify a higher speed.

Source: Snehal_NEXORA_FogNe.pdf

Driver HMI

■ SLOW DOWN

SAFE SPEED: 8 km/h

CLEAR: 18 m REQUIRED: 25 m

TRUCK AHEAD: 18 m

Source: PrathameshFOG_SHIELD_X.pdf • Snehal_NEXORA_FogNe.pdf

7. What happens when something fails?

Failure	System response
Camera fails	Radar remains the local safety path.
V2V fails	Local vehicle safety continues.
Road Guardian fails	Vehicle local safety continues.
Internet/cloud fails	Local safety continues.
GNSS becomes uncertain	Use route graph + IMU and conservative local logic.
Hazard message becomes stale	Reject it rather than treating it as current evidence.
Local radar detects hazard	Peer messages cannot clear the local hazard.

Source: AboliFogNet_4.1.pdf • Chatgpt_FogNet_Nexus.pdf • Snehal_NEXORA_FogNe.pdf

Desian rule

Network loss must make the system more conservative, not blind. New evidence can reduce the safe operating envelope immediately; it should not silently increase it.

Source: Snehal_NEXORA_FogNe.pdf

8. Prototype implementation

Vehicle prototype

Component	Role
ESP32 / STM32	Local controller
Radar development sensor	Obstacle distance
Camera	Object classification
IMU	Vehicle motion
GNSS	Position / route context
Buzzer + vibration + display	Driver warning
Laptop / Raspberry Pi	AI + dashboard for prototype

Roadside + blind curve prototype

Road Guardian: ESP32 + visibility sensor + temperature/humidity + local wireless + battery.
Virtual Block: two wireless nodes with indicators representing the blind segment.

Source: AboliFogNet_4.1.pdf • Snehal_NEXORA_FogNe.pdf • Chatgpt_FogNet_Nexus.pdf

Prototype demo sequence



9. Feature-to-source map

The table below identifies the main uploaded document(s) used for each feature in the final synthesis. Some features are intentionally combined or modified rather than copied directly.

Our feature	Main source document
Safe Corridor	Snehal_NEXORA_FogNe.pdf
Dynamic Safety Bubble	AboliFogNet_4.1.pdf
Radar-first sensing	Snehal_NEXORA_FogNe.pdf
Radar + Camera	PrathameshFOG_SHIELD_X.pdf + GLM_Fogswam.docx
Local safety during network failure	AboliFogNet_4.1.pdf
V2V cooperative perception	GLM_Fogswam.docx
Hazard Capsule	GLM_Fogswam.docx + Snehal_NEXORA_FogNe.pdf
Predictive hazard sharing	PrathameshFOG_SHIELD_X.pdf
Virtual Block	Snehal_NEXORA_FogNe.pdf
Movement Token	Chatgpt_FogNet_Nexus.pdf
Road Guardian	AboliFogNet_4.1.pdf
Fog trend	Chatgpt_FogNet_Nexus.pdf + Snehal_NEXORA_FogNe.pdf
BrakeTwin	Snehal_NEXORA_FogNe.pdf
Physics-based safe speed	AboliFogNet_4.1.pdf + Snehal_NEXORA_FogNe.pdf
AI-assisted perception	GLM_Fogswam.docx + PrathameshFOG_SHIELD_X.pdf
Driver warning	PrathameshFOG_SHIELD_X.pdf + Snehal_NEXORA_FogNe.pdf
Control-room dashboard	Chatgpt_FogNet_Nexus.pdf + AboliFogNet_4.1.pdf
Safety Black Box	Chatgpt_FogNet_Nexus.pdf
Failure/degraded modes	AboliFogNet_4.1.pdf + Snehal_NEXORA_FogNe.pdf

10. Final team statement

FogNet Sentinel combines the strongest compatible ideas from the uploaded solutions into one practical architecture:

RADAR-FIR; SAFBRACKETWIV2VIRTUAL BLOCK
LOCAL SAFCORRIDOR HAZARD BAND CURVE

The central engineering idea is simple: the vehicle should not continue at a speed that requires more stopping distance than the road ahead can currently be trusted to provide.

The project is designed as an advisory prototype. Any production vehicle intervention such as automatic braking would require appropriate vehicle integration, hazard analysis, validation and certification.

Source: Snehal_NEXORA_FogNe.pdf

Uploaded source documents

- Chatgpt_FogNet_Nexus.pdf
- Claude_FogNet_Alive.pdf
- AboliFogNet_4.1.pdf
- GLM_Fogswam.docx
- PrathameshFOG_SHIELD_X.pdf
- Snehal_NEXORA_FogNe.pdf